

The Alberta Buck – The BasketWheel (prototype)

How the BuckBasket Earns: Its Credit, Its Own Arbitrage, Its Timing, and the Wheel
That Turns Them

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The BuckBasket says the basket earns from tolls, from correcting its own prices, and from timing slow swings. This paper is the machinery and the arithmetic behind those three claims.

A pool of liquidity is a rebalancer that trades at stale prices. In a market that reverts, the arbitrageur who re-pins it keeps nearly all of the rebalancing premium, and the pool keeps only its fees. The equity basket takes that premium back with *consistency cycles*: three swaps round a commodity's three prices, needing no capital and unable to lose. It times the slow swings a pool cannot harvest with a *director* that leans against a commodity's excursion from its long memory, but only once the run has turned. A *monetary desk* that trades BUCK against the basket runs beside it, never inside it, in an account of its own. All of it runs on a *work wheel*: a bounded state machine that anyone may advance and that pays its callers from what it captures.

This is a prototype. The contracts exist and run in a simulated economy and in a public sandbox. None of it is deployed or audited, and the numbers below are a simulation's. (PDF, Text)

Contents

1	Status	2
2	The machine	2
3	Where a pool's harvest goes	3
4	The consistency cycle	3
5	The director: timing the tides	5
6	The desk: a separate machine	6
7	The work wheel	6
8	What the simulations say	7
9	Open questions	8
10	Risks	9
11	Conclusion	9

1 Status

Piece	State (2026-09-30)
The equity basket (<code>src/basket/</code>)	built: shell, components facet, venue facet, turn director; Foundry-test
The monetary desk (<code>EquityDesk</code>)	built as its own credit holder; an invariant suite fuzzes the separation
The work wheel (<code>src/wheel/</code>)	built: one engine, instantiated per machine with that machine's kinds
The consistency cycles (<code>ArbKind</code>)	built; 16 Foundry tests on real Uniswap V3 pools
In the simulated economy	running; the savings sandbox runs it live
The two-year experiments on this basket	running (the earlier ones ran on the pro-rata basket it replaced)
Deployment, audit	none

Read every figure below as "in a simulated economy", not as a forecast. The contract surface is in the Ethereum implementation; the design record, with every ruling and measurement, is doc/BASKET-EQUITY.org and doc/BASKET-WHEEL.org.

2 The machine

The basket is shares of equity, one lien, and a wallet.

```
gross(mark)   = sum over pools ( 2 L_i sqrt(P_i)  +  idle TOKEN_i x P_i )
equity(mark)  = gross(mark) + signedBalance(basket) + reliefAccrued(basket)
share price   = equity / shares
```

$2 L \sqrt{P}$ is a full-range position's fair value at a price. It is computed from the pool's liquidity and a price, never from token balances, so tokens sent to a pool cannot inflate it. There are three marks: the TWAP, HIGH (each pool at the higher of spot and TWAP) and LOW (the lower). A deposit is valued at LOW against a basket at HIGH; an exit at LOW. Each side pays a charge the size of the swap the wheel will make for it: $(1 - K)/2$ of the pool fee for TOKEN in, $(1 + K)/2$ of the dearest pool's fee for BUCK in or out. Nobody can enter or leave at a price that moves value between the holders.

The credit. The basket is an ordinary BUCK account holding one self-issued *marked* credit, marked before anything that spends at `equity(LOW)` – the depositors' equity, and nothing else. Buck therefore enforces

```
lien(basket)  <= K x equity(LOW)
```

on every draw. The basket never mints or burns: it spends past its held BUCK, which draws on the credit, and BUCK it receives repays the lien. The lien earns Jubilee relief like any other. A cut in K changes no lien; an under-water account simply cannot draw. There are no margin calls: the excess is repaid only by the wheel's paced Trim.

Exits. The treasury takes 25% of a receipt's gain over its cost basis, as shares, and nothing of a loss. The rest is paid in BUCK when Buck lets the basket spend it. Otherwise it is paid in kind: the exit's fraction f of every position and of the idle TOKEN, by transfer, its share of the lien repaid from the BUCK its positions return. No swap is forced on anyone.

The wallet absorbs the flows. Its target liquidity is the larger of 1% of the gross and a flow-sized amount – enough that half of it covers two standard deviations of the daily net flow times $1 + K$

– capped at 20% of the gross. Deposits and exits net against each other inside it, and an ordinary exit touches no pool.

3 Where a pool’s harvest goes

A full-range position is a constant-mix rebalancer that trades at stale prices. It sells TOKEN as the price rises and buys as it falls – the reversion harvest – but it does so at its own price, which trails the market. Whoever moves it to the market price keeps the difference. That difference has a name, *loss-versus-rebalancing* (Milionis, Moallemi, Roughgarden and Zhang, 2022), and a size. For a constant-product pool of value V whose asset moves with volatility σ :

$$\text{LVR} \sim V \times \sigma^2 / 8 \quad \text{per unit time, before fees}$$

In a reverting market that is almost exactly the rebalancing premium. So the harvest goes to the arbitrageur, and the pool keeps the fees on the arbitrageur’s volume. For one pool reverting with a 20-day half-life at 1.5% a day, the premium is 1.0% a year of the position:

outside arbitrage	pool fee	the pool keeps	the basket’s own arbitrage keeps
once a day	0.3%	0.22%	0.90%
once a day	1%	0.49%	0.90%
every hour	0.3%	0.52%	0.75%
every hour	1%	0.79%	0.83%

The finer the trading against the fee, the more the fee recovers by itself; the coarser, the more only the basket’s own arbitrage can.

A pool cannot harvest slow swings at all. A constant-mix rebalancer earns the variance of short-horizon moves – the quadratic variation. A smooth cycle has almost none, however large its amplitude. A swing is harvested only by trades timed across pools: selling one leg near its top, buying another near its bottom.

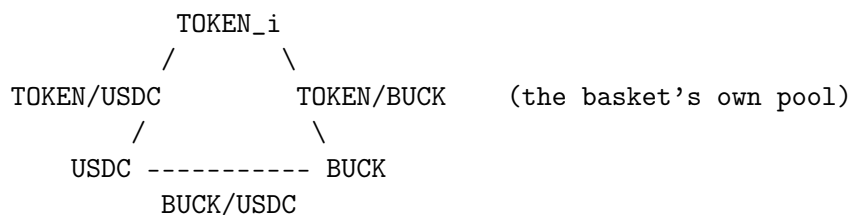
So a basket’s excess return over holding its commodities has three sources, each with its own harvester:

mispricing	harvester	its risk
venue inconsistency (a stale TOKEN/BUCK pool)	the consistency cycle	none on a cycle
relative value (a leg rich against its past)	the director’s lean	whipsaw in a secular trend
BUCK against its basket	the monetary desk	inventory that persists

Beneath all three are the tolls: the fees on the basket’s own pools, which the Sync step collects.

4 The consistency cycle

Three pools quote each constituent:



If the three prices agree, going round the triangle returns what went in, less the fees. If they disagree, one direction returns more. Going round that way, by the right amount, closes the gap and keeps the difference.

4.1 The algebra

A constant-product leg with reserves (X, Y) and fee factor g (one minus the fee) pays, for an input x ,

$$\text{out}(x) = g Y x / (X + g x) = r x / (1 + k x), \quad r = g Y / X, \quad k = g / X$$

a Moebius map through the origin: r is the leg's marginal rate at zero size, k its curvature (the inverse of its depth). Such maps compose into maps of the same form, so a whole cycle is two numbers:

$$\begin{aligned} r &= r_1 r_2 r_3 \\ k &= k_1 + k_2 r_1 + k_3 r_1 r_2 \\ \text{cycle}(x) &= r x / (1 + k x) \end{aligned}$$

$$x^* = (\sqrt{r} - 1) / k \quad \text{profit}^* = (\sqrt{r} - 1)^2 / k$$

The cycle pays exactly when $r > 1$, and the optimum closes the gap down to the fees, no further. A full-range V3 position is constant product on its virtual reserves, so the closed form is exact for the basket's own pools and a good approximation elsewhere. `CycleMath` in `src/wheel/ArbKind.sol` is this section in Solidity.

4.2 Why it cannot hurt

- **No capital.** A V3 pool pays a swap's output before it collects the input, in a callback. The first leg pays out; the other two run inside its callback; what they return pays the first. No inventory, no flash loan, no flash mint. USDC exists only between two swaps inside one transaction.
- **No change in supply.** No BUCK is issued or destroyed; value moves from the venues to the basket. A cycle never trades BUCK against the basket's value – that is the desk's business.
- **It cannot lose.** The cycle checks its realized output against its input and reverts below a minimum edge. Someone who pushes a pool first only pays the basket to push it back.
- **It keeps the gauge true.** The controller steers on the basket's value in BUCK, read from these pools. Closing their gaps makes that reading follow the commodities instead of lagging them by an arbitrageur's threshold.

The same three swaps, in the same rotation, earn the same whichever token the cycle starts from; the start decides only what the profit is made of. Started from the TOKEN, the profit lands in the basket's wallet, and every share is worth more. Started from BUCK, it funds the wheel's callers' reserve (below) up to a cap, and the rest goes to the basket's account.

4.3 Who can take which gap

A searcher going round the triangle pays three pool fees. It is tempting to think the basket pays only two – it is nearly the only liquidity in its own pool, so the fee it pays there comes back – and so can take a band of small gaps nobody else can. It cannot. On its own share of a pool the basket is trading with itself: at the outside price, a "capture" inside the outsiders' band gains only the outside liquidity's share of the gap, and pays the outside cost on all of it. That is a rebalancing bet, not an arbitrage, and measured it gains nothing (within 0.02-0.16% a year, either sign).

What the basket *can* do is take the outsider's trade first. The gaps worth taking are the ones wider than all three fees, and on today's Uniswap V3 the basket must race searchers for them. Winning that race by construction needs control of the order of swaps: a Uniswap v4 hook that runs the cycle before the first swap of each block, or an auction for the right to go first (the auction-managed AMM of Adams, Moallemi, Reynolds and Robinson, 2024). That is the upgrade path.

5 The director: timing the tides

`EquityTurnDirector` (optional) keeps, for each leg, a ladder of exponential moving averages of the pool's TWAP against the basket's mean:

- **The lean.** Each declared weight is scaled by `exp(-tilt x excursion)` and the weights renormalized, the excursion being the leg's deviation from the basket's mean against a long *anchor* EMA (1280 days). A leg rich against its own past is leaned down. `tilt` defaults to 1.
- **The turn.** Six shorter EMAs, 5 to 160 days, each vote when moving back toward the anchor. A quorum of votes (default 4 of 6) is a turn. A leg not turned is still running, and a run is let run: the director never trims a leg still rising or buys one still falling.
- **The leash.** A leg more than 30% off its *declared* weight trades regardless of the gate. The lean times the mandate; it never replaces it.

The Fund step buys the director's pick; the Trim step sells it. Proceeds wait in the wallet between a sale and the buy that is due, so the director never has to buy a falling knife to stay invested.

Measured on synthetic worlds (below), the turn gate alone adds little on pool positions – a pool's weight moves by only half its price move, and the pool already sells into every run. The value is in the lean. The gate is the regime knob:

over the arbitrage alone, a year (tilt 1 / 2)	slow cycles	fast reversion	random walk
ungated	-0.1 / +0.3	+2.7 / +4.5	0.0 / -0.3
quorum 4 of 6	+1.9 / +3.7	+0.2 / +0.5	-0.1 / -0.4

In momentum cycles the gate is what makes the lean pay; in memoryless reversion it costs, by trading late; in a walk every setting is noise. The longer the anchor, the better the lean (320 days: +1.1%; 1280 days: +1.6%; all but fixed: +1.9%, at tilt 1 in the cycle world). An anchor's risk is a permanent repricing, which the leash bounds.

6 The desk: a separate machine

The third mispricing, BUCK against its basket, is the monetary desk's: it issues BUCK when BUCK is dear and absorbs it when cheap, in the four quadrants of Basket Operations. Beside the equity basket it is `EquityDesk`, and it is a separate contract with its own account at Buck, its own marked credit (marked at its own net value: its `TOKEN` at the low marks, plus its account and the relief accrued on it), its own founding grant, and its own work wheel. From the basket it reads two things: the basket's equity, which sizes its bounds, and the basket's constituents, so it trades in the same pools through the same venue facet.

- D1: the desk spends only after a mark at its own net value, so Buck holds its lien within $K \times$ that value.
- D2: nothing the desk holds or owes enters the basket's books, mark or limit.
- D3: the relief on the desk's lien is the desk's, and melts its outstanding issuance.

The rule behind it: Buck keys everything by address – one balance, one lien, one limit per account – so bookkeeping sub-accounts inside one contract are invisible to it. The desk was first built as one, and in the savings sandbox its founding grant quietly raised the depositors' limit: a lien of 6.3M on 5.8M of equity at $K = 0.75$. Two machines now means two contracts, two accounts, two credits and two wheels. A fuzzing suite (`test/basket/EquityDeskInvariant.t.sol`) checks the wall after every call; inflating the basket's mark by 2% makes it fail.

7 The work wheel

Someone has to notice the work and send the transaction. The basket does not wait for a keeper paid by the hour: its work runs on a *work wheel*, a bounded state machine any caller advances with `tick(maxWork, maxScan)`, which does each block's work once however many callers there are.

- **Slots over task kinds.** Each kind of work owns a range of slots, and a cursor walks them round robin. The equity basket's components are $2N + 3$ slots: Daily (the flow estimate, the relief, the director's sample), Sync and Deploy for each of N pools, Fund and Trim. The consistency cycles are one slot per triangle. The controller's `compute` and the director's refresh are kinds too. Kinds are Solidity mixins over a common chassis, so a wheel is assembled from the kinds it wants, and a kind with no target has no slots. One engine, `BasketWheel`, is instantiated per machine: the basket's wheel carries the basket's kinds, the desk's carries only the desk's.
- **Each step is small.** Every component step marks the credit first, and trades at most 1% of a pool's depth, so the outside arbitrage re-pins between steps and no step moves a price far.
- **Amortizing the scan.** Deciding whether an arbitrage slot is due costs three cold pool reads, about 14,000 gas. A call examines at most `maxScan` slots, the cursor moves past every slot examined, and the block is marked idle only once consecutive calls have found every slot not due. After that a further call is a storage read and a return. Any trade that touches the basket re-arms it.
- **A clock per chain.** The wheel keys its work to the chain's own block number; where that is not the chain's own (Arbitrum reports the L1's), a small adapter supplies the right one.

Folded into a transaction a caller sends anyway, a tick whose block is done costs a few thousand gas. A working tick costs what its work costs: about 365,000 gas for a cycle.

7.1 Paying the caller

The goal: the calling pays for itself on every chain, without the basket knowing the gas price or the price of ETH.

- **No cure, no pay.** A tick pays only for work done. Callers simulate first and send only when there is work; an idle call is the caller's loss, never the basket's.
- **A share of the value.** A kind that captures value pays its caller a share s of it. A caller calls when $s \times \text{value}$ exceeds its gas, so the action margin sets itself per chain: on an L1 the wheel acts on large gaps, on a rollup on small ones.
- **A reserve for the work that captures nothing.** Upkeep earns nothing by itself. A slice of the yield flows into a reserve, and each working tick pays its caller $kappa$ of the reserve's balance. The reserve builds when the wheel is under-called and pays less when it is over-called; at equilibrium a working tick pays the funding rate over the calling rate, and callers call until that equals their gas. The calling rate sets itself, with no oracle. A cap sends the excess to the basket: the reserve is a gas budget, not a second yield.
- **Competition is harmless.** Callers racing for a tick bid their priority fees up to the pay less their gas; that transfer is the block builder's, out of the pay. The basket's cost is fixed at s and $kappa$.

8 What the simulations say

Synthetic worlds (`alberta_buck/sim/basket_harvest.py`): independent TOKENs at about 1.5% a day in three regimes – fast reversion (a 10-day half-life), slow damped cycles with momentum, and a random walk as the control – with an outside arbitrageur re-pinning every pool each step. Eight seeds over four years; a year's return over simply holding the starting commodities ("pools alone" is the basket with no arbitrage of its own):

regime	pools alone	+ the basket's arbitrage	+ the lean, turn-gated
fast reversion	+0.66%	+1.63%	+1.88%
slow cycles	+1.09%	+1.52%	+3.24%
random walk	-1.19%	+0.55%	+0.35%

The basket's arbitrage recaptures what the outsiders took, to the basis point. Costless, perfect rebalancing at the market price would have earned 2.6% in the fast world: pools alone keep about a quarter of it, and the arbitrage lifts that to about two thirds. With realistic flows – two arrivals a day, sixty-day holds – a long-term holder earned about 3% a year, because deposits and exits net inside the wallet and an exit touches no pool.

A simulated economy (`alberta_buck/sim/`): the real contracts on an in-process EVM, six commodity TOKENs on their 2020-2025 price histories, the controller, depositors, savers, debtors and a cast of arbitrageurs. These runs predate the equity basket: the cycles ran on the pro-rata basket it replaced.

- The cycles captured 2-3% of the basket's value a year in the first calm months, and about 1.3% a year over two years; their calls paid for themselves even at 5-gwei L1 gas. About half came from gaps the outside arbitrageurs would have closed.
- The gauge became truer: the basket's value in BUCK, against its value at the commodities' reference prices, erred a third less at the median and 43% less at the 90th percentile.
- Through two years of shocks – demand and supply at one and four times a whale's budget, slow grinds, cohorts of debtors, a book-loading attack, a liquidity exit – nothing broke, the gauge was truer on every run, and the wheel transmitted shocks without amplifying them: BUCK's dollar price swung less, while the basket's pools, which now followed the market, took part of each dislocation and were paid for it.

Two cautions. The simulated arbitrageurs demand a 60-basis-point edge over the fees, so they leave wider gaps than competitive searchers would; against a real market the wheel's take will be smaller. And the two-year experiments are being rerun on the equity basket and the corrected Jubilee accounting now.

9 Open questions

1. **A competitive market.** How much remains when real searchers compete the large gaps down to gas, and does thinning their participation widen the gaps beyond the wheel's reach?
2. **The large gaps.** A v4 hook, or an auction for the right to go first, would capture what v3 must race for; each has its own trust and liveness questions.
3. **Leverage through a K cut.** Drawing the full K on each deposit keeps the basket more levered through a cut (a lien of 0.92 of equity at the stress peak, against 0.80), with returns within noise. The paced Trim is the only repayment, by design; whether its pace is right is open.
4. **Depositors as shock absorbers.** Pools that follow the market carry BUCK's moves into the depositors' positions. They were paid for transient shocks; under a persistent increase in supply their realized profit on the calm runs fell to nothing. Whether the wheel or the cast did that is not yet separated.
5. **A residual oscillation.** With the agents' decisions randomized, BUCK's dollar price is white noise day to day, but the basket's value and K still reverse direction more often than noise would (about 75% of days against 67%). Some loop still overshoots; it predates the equity basket.
6. **The desk's signal.** An earlier desk read BUCK's deviation from an equal-weighted average of each pool's move since the director first looked, and sat one-sided at its cap for two years. The proposed remedy is to read the basket's own value against par.
7. **The share and kappa,** and where the callers' reserve is funded from: fixed, per chain, or adaptive to the reserve and the calling rate.
8. **Cycle sizing.** v1 caps a cycle at a fraction of its first leg's reserve; every leg should be capped.
9. **Depth.** A thin BUCK/USDC pool caps every cycle. The basket owns no BUCK/USDC position, and should not; who provides that depth is a question for any deployment.

10 Risks

- **The callbacks and the credits.** The swap callback must accept only the pool it expects, only inside its own cycle; the cycle entry only from the wheel itself; the basket's credit entry points only from its wheel. The contracts do all three and test them, unaudited.
- **A farmable reserve.** The share s cannot be farmed: pushing a pool costs more than closing the gap returns. $kappa$ per working tick is not tied to the work's size, so a large reserve might reward many small, cheaply opened gaps. The minimum edge, the reserve's cap and a pay bounded by the work's value are the defences to test.
- **Displacing the arbitrageurs.** If external arbitrageurs thin out, the gaps beyond the wheel's reach may close more slowly.
- **Gas on an L1.** A working tick is expensive; on an L1 only large gaps pay, and those are the ones searchers race for. The case is strongest on rollups.
- **Liveness.** No calls, no harvest, and new deposits wait in the wallet. Nothing already placed is at risk, and exits are paid from the wallet or in kind.
- **A trending market.** None of this changes the basket's exposure. In a secular trend a constant-mix basket still trails a buy-and-hold of the runaway; the director's leash and the admission rule are the only defences (the asset that never comes back).

11 Conclusion

The first BuckBasket assumed that the world's arbitrageurs keep it honest and pay it to be kept. The first half is true. The second is only partly true, because the price of keeping a pool honest is paid by the pool. The equity basket brings that work home: it closes its own gaps with cycles that need no capital and cannot lose, times the tides its pools cannot feel, keeps the desk's books out of its own, and lets anyone who finds it worth their gas turn the wheel. If the early readings survive a competitive market and an audit, the savings basket earns a little more – and its valuation tells the controller the truth a little sooner.

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